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ChE 210

Quiz #1

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Experimental data for a liquid/vapor binary mixture of pentane (P) and heptane (H) is given in the table. The quantity z_{liq} gives the mole fraction of pentane in the liquid phase while z_{vap} gives the mole fraction of pentane in the vapor phase. The data can be linearized using the relation $Y = A X$ where A is the separation factor, $X = z_{liq}/(1 - z_{liq})$, and $Y = z_{vap}/(1 - z_{vap})$.

For a general straight line with the intercept equal to zero $y = a x$, the "best fit" slope for perpendicular offsets is given by $a = -\bar{B} \pm \sqrt{\bar{B}^2 + 1}$ where $\bar{B} = \frac{S_{xx} - S_{yy}}{2S_{xy}}$ and the

averages are defined as $S_{xx} = \frac{1}{n} \sum_{i=1}^n x_i^2$, $S_{yy} = \frac{1}{n} \sum_{i=1}^n y_i^2$, and $S_{xy} = \frac{1}{n} \sum_{i=1}^n x_i y_i$, respectively. If $S_{xx} - S_{yy} > 0$, use the plus sign in front of the square root in the expression for the slope. If $S_{xx} - S_{yy} < 0$, use the minus sign.

(a) Give the units on the mole fractions z_{liq} and z_{vap} .

(b) Give the units on the mole fraction ratios X and Y .

(c) Calculate the "best fit" value of the separation factor using perpendicular offsets as described above. Show all of your work.

Mole Fraction z_{liq}	Mole Fraction z_{vap}
0.867	0.984
0.594	0.925
0.398	0.836
0.254	0.701
0.145	0.521
0.059	0.271

(a) $z_{liq} = \left[\frac{\text{mol}_{liq}}{\text{mol}_{tot}} \right]$ $z_{vap} = \left[\frac{\text{mol}_{vap}}{\text{mol}_{tot}} \right]$

$\therefore y = -0.6625x$ is best fit line of equation, by $\perp$ offset.

(b) $x = \left[\frac{\text{mol}_{liq}}{\text{mol}_{tot} - \text{mol}_{liq}} \right]$ $y = \left[\frac{\text{mol}_{vap}}{\text{mol}_{tot} - \text{mol}_{vap}} \right]$

(c)

X	$Y = (z_{vap}/1 - z_{vap})$ $X = z_{liq}/1 - z_{liq}$
6.519	61.5
1.463	12.3
0.661	5.098
0.340	2.344
0.170	1.088
0.062	0.372

$S_{xx} = 14.2$ $S_{yy} = 11140$ $S_{xy} = 70.626$

$S_{xx} - S_{yy} < 0$, $\bar{B} = \frac{14.2 - 11140}{2 \cdot 70.626} = -7.97$

$a = -\bar{B} - \sqrt{\bar{B}^2 + 1} = -0.6625$

